

8 (short)

(a) A certain memory chip has 11 address lines and 8 data lines. It is connected to a microprocessor with a 16-bit address bus and an 8-bit data bus. Determine the capacity of the memory chip in bytes. [2]

(b) Determine a Boolean expression for the $\overline{CS}$ input such that the memory chip appears in the microprocessor's memory map between locations $D800_H$ and $DFFF_H$. Draw a circuit showing how the memory chip should be connected to the microprocessor. [8]